

PROJECT REPORT

ON

**HIGH RESOLUTION SPECTROSCOPY OF
CLASSICAL CEPHEIDS**

Submitted in partial fulfillment of the requirements for the award of the degree of

MASTER OF SCIENCE

IN

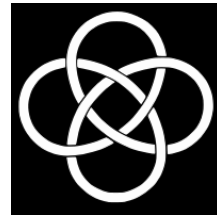
PHYSICS

Submitted by

Uvais Ul Qarni S

Department of Physics

University of Calicut



Under the Guidance of

Dr. Anupam Bhardwaj

Assistant Professor

Inter-University Centre for Astronomy and Astrophysics (IUCAA)



अंतर-विश्वविद्यालय केंद्र : खगोलविज्ञान और खगोलभौतिकी

IUCAA

Inter-University Centre for Astronomy and Astrophysics

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Dr. ANUPAM BHARDWAJ - Assistant Professor - anupam.bhardwaj@iucaa.in +91-20-25604-204

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This is to certify that the work entitled "**High-Resolution Spectroscopy of Classical Cepheids**" submitted by **Mr. Uvais Ul Qarni S** (CUAYPHY019), in partial fulfillment of the requirements for the award of the degree of Master of Science in Physics from the Department of Physics, University of Calicut, Kerala, is a bonafide record of research work carried out under my guidance and supervision.

The project work was undertaken at the Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, during the period from February - May 2026.

Dr. Anupam Bhardwaj
Assistant Professor
IUCAA, Pune

Mailing Address : IUCAA, Post Bag 4, Ganeshkhind, Pune - 411 007, India.

Location : Meghnad Saha Road, Savitribai Phule Pune University Campus, Pune - 411 007.

Phone : 020-2569 1414, 2560 4100 **I Fax :** +91-20-2560 4699

E-mail : root@iucaa.in

Uniform Resource Locator : <http://www.iucaa.in>

DECLARATION

I, **Uvais Ul Qarni S**, hereby declare that the project report entitled “**High-Resolution Spectroscopy of Classical Cepheids**” submitted to the **University of Calicut** in partial fulfillment of the requirements for the award of the degree of **Master of Science** in Physics, is a record of the original work done by me under the supervision and guidance of **Dr. Anupam Bhardwaj** at the Inter-University Centre for Astronomy and Astrophysics (IUCAA).

I further declare that this work has not been submitted, either in full or in part, for the award of any other degree, diploma, fellowship, or similar title at any university or institution.

Place: Malappuram

Date:

Uvais Ul Qarni S
Department of Physics
University of Calicut

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ABSTRACT

Classical Cepheids (DCEPs) are considered to be the most crucial and widely-used distance ladders to calibrate Galactic and extragalactic distances. The period-luminosity (PL) and period-Wesenheit (PW) relations help us for this determination. Precise estimates of atmospheric properties and chemical compositions of these stars allow researchers to explore metallicity dependence of the mentioned relations and obtain useful insights into the chemical evolution of the Galactic disc. Here, I report the results of a spectroscopic analysis of one such star—Classical Cepheid SV Persei—based on observations and data obtained from the high-resolution echelle spectrograph ESPaDOnS at CFHT.

Equivalent widths of chosen Fe I and Fe II lines from an available line-list were measured with a tool called Automated Routine for line Equivalent Widths in stellar Spectra (ARES). The entire spectral analysis was automated and optimized in Python. I utilized the pymoog wrapper to execute the MOOG radiative transfer code, performing the abundance analysis under the assumption of Local Thermodynamic Equilibrium (LTE) using 1D Kurucz model atmospheres.

Effective temperature, surface gravity, and micro-turbulent velocity of SV Persei were determined from excitation equilibrium, ionization equilibrium, and iron abundance independence conditions. These converged atmospheric parameters were then used to determine the metallicity of the star. Synthetic spectra were constructed and compared to the observed spectrum in order to check consistency between measured parameters and spectral data.

This automated analysis demonstrated its potential and usefulness for studying Cepheid variables' parameters.

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Chapter 1

Introduction

1.1 Variable Stars

Stars with changing magnitudes are called *variables*. The variation in their luminosity can occur over timescales ranging from fractions of a second to several years. The light curve, which is a plot of brightness versus time, reveals the period, amplitude, and shape of the variation, which in turn helps us understand the underlying physical mechanisms of such stars.

Variables are typically classified into two types [16]:

1. Intrinsic variables
2. Extrinsic variables

1.1.1 Intrinsic Variables

For intrinsic variables, changes in brightness result from changes occurring either within the stellar interior or at its surface. These stars are classified based on the physical mechanisms driving this variability.

Pulsating Variables

These stars physically expand and contract. As the radius and surface temperature change, so does the luminosity. Pulsations are driven by mechanisms such as the κ -effect (changes in opacity in ionisation zones) within a partially ionised layer [38]. Major subgroups include:

- Cepheids
- W Virginis Stars

- RR Lyrae Stars
- Mira Variables
- Semiregular and Irregular Variables

Eruptive Variables

Eruptive variable stars exhibit unpredictable fluctuations in brightness due to material either accreted to or lost from the star. Despite their name, these are not the same as cataclysmic variables because non-thermonuclear mechanisms create the eruptions [31]. Subgroups include:

- T Tauri Stars
- Flare Stars
- Luminous Blue Variables

Cataclysmic Variables

Cataclysmic variables (CVs) are binary stars that contain a white-dwarf primary and a mass-transferring secondary. The stars are so close together that the white dwarf's gravity distorts the secondary. The white dwarf accretes matter from the companion star, which is typically a late-type, low-mass main-sequence star. The secondary is thus often called the donor star and is usually less massive than the primary [20]. Subgroups include:

- Novae
- Dwarf Novae
- Recurrent Novae
- Supernovae

1.1.2 Extrinsic Variables

Extrinsic variables are those in which the light output varies either due to the rotation of the star or due to a process external to the star itself. The two main types of extrinsic variable stars are eclipsing binaries and rotating variables.

Eclipsing Binaries

In a binary star system, two stars orbit each other under their mutual gravitational pull. When one star passes in front of the other, the light from the eclipsed star is completely or partially blocked from our line of sight, producing a periodic variability in the combined light curve.

Rotating Variables

These are single stars whose surfaces are not uniform, either due to starspots (cool, dark regions similar to sunspots), chemical abundance patches, or strong magnetic fields that concentrate certain elements on the surface. As the star rotates, the non-uniform surface moves in and out of view, causing a smooth periodic brightness variation.

1.2 Classical Cepheids

Classical Cepheids (DCEPs) are a class of radially pulsating yellow supergiant stars that occupy the Cepheid instability strip in the Hertzsprung–Russell (H-R) diagram [17]. They are among the most important objects in observational astrophysics, serving simultaneously as laboratories for stellar pulsation physics, tracers of Galactic chemical evolution, and primary distance calibrators in the cosmic distance ladder. Their defining characteristic is a strict relationship between pulsation period and luminosity—the period–luminosity (P-L) relation—which has made them indispensable tools for measuring extragalactic distances for over a century.

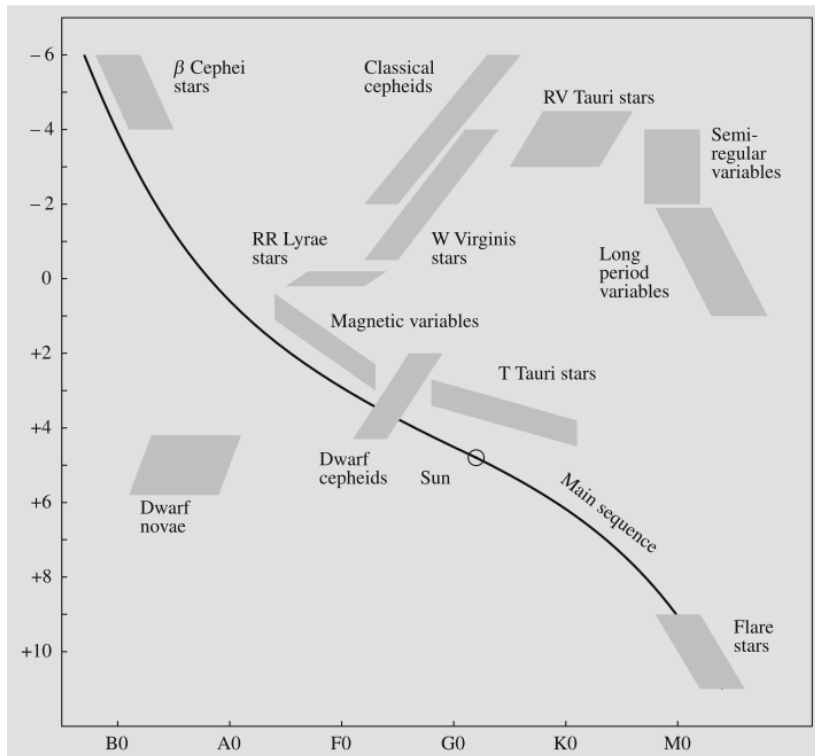


Figure 1.1: Locations of various classes of variable stars in the Hertzsprung–Russell diagram. The x-axis represents the spectral type (effective temperature increasing to the left), and the y-axis represents the absolute magnitude (M_V , luminosity increasing upwards). (Credit: [17]).

The prototype of the class, δ Cephei, was first observed to vary in brightness by John Goodricke in 1784 [26]. He correctly attributed the variability to a periodic physical cause [14, pp. 4–5], although the interpretation of pulsation came much later.

The real breakthrough came in 1912, when Henrietta Swan Leavitt, working at the Harvard College Observatory, made a systematic study of variable stars in the Small Magellanic Cloud (SMC) [21]. Since all stars in the SMC are roughly the same distance from Earth, differences in their apparent magnitude directly reflected differences in absolute luminosity. Leavitt found that the brighter Cepheids had longer periods, establishing the first period–luminosity relation. This discovery transformed Cepheids from astrophysical curiosities into standard candles.

1.3 The κ -Mechanism

Classical Cepheids are pulsating stars whose pulsations are driven by the κ -mechanism (kappa mechanism) operating in the partial ionisation zones of helium in their stellar envelope. The physical process can be described in three steps:

During compression (inward motion of the photosphere): The helium ionisation zone ($\text{He II} \rightarrow \text{He III}$, at $T \approx 40,000$ K) is compressed and becomes more opaque (higher opacity, κ). The layer traps the energy rather than efficiently radiating it outwards, leading to increased heating and a buildup of pressure.

During expansion (outward motion of the photosphere): The stored energy is released, causing the outer layers to expand outwards and amplifying the motion. It functions as a thermodynamic heat engine: energy is absorbed during compression and released during expansion, acting as the reverse of a normal damping layer.

Self-sustaining cycle: As long as the ionisation zone is situated at the correct depth within the stellar envelope (neither too deep nor too shallow), the mechanism sustains and amplifies the pulsation against radiative damping. Consequently, only stars within a specific temperature range (the instability strip) pulsate, as the ionisation zone must be at the precise optical depth.

A secondary contribution comes from the hydrogen ionisation zone, which is also involved in driving the pulsation, especially in cooler Cepheids. The growth rate, period, and amplitude of the pulsation depend on the delicate balance between these driving layers and the damping occurring in the deeper envelope.

1.4 The Leavitt Law (Period–Luminosity Relation)

The P-L relation (also called the Leavitt Law) is the cornerstone of Cepheid-based distance measurements [24]. In its modern form:

$$M_V = \alpha \cdot \log P + \beta \quad (1.1)$$

where M_V is the absolute visual magnitude, P is the pulsation period in days, and α and β are empirically determined coefficients. The relation becomes tighter (exhibiting smaller scatter) in near-infrared bands (H , K) where the effects of dust extinction and temperature spread are minimised, making infrared P-L relations the preferred tool for extragalactic distance work.

Metallicity dependence: The P-L relation contains a metallicity term [29] — at a given period, more metal-rich Cepheids are slightly brighter (or slightly fainter, depending on the band and the calibration source). The magnitude and sign of this correction remain a topic of active research and serve as a primary motivation for chemical abundance surveys of Galactic Cepheids, such as the C-MetaLL survey [37].

The Hubble constant tension: The calibration of the Cepheid P-L relation is central to local measurements of the Hubble constant, H_0 . The current tension between the local distance-ladder value of H_0 ($\approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ from HST Cepheids + Type Ia supernovae) and the early-Universe value derived from the cosmic microwave background ($\approx 67 \text{ km s}^{-1} \text{ Mpc}^{-1}$ from Planck) is one of the outstanding problems in modern cosmology. Improving Cepheid P-L calibrations, specifically the metallicity correction, is a key pathway towards resolving this discrepancy.

1.5 Pulsation Modes and Classification

There are two primary modes in which classical Cepheids pulsate:

Fundamental mode (F): The entire stellar envelope expands and contracts as a single unit. These are the most prevalent classical Cepheids, typically exhibiting periods between 1 and 100 days. They follow a well-defined P-L relation and display larger amplitudes ($\Delta V = 0.3\text{--}1.5 \text{ mag}$).

First overtone (1O): The envelope oscillates with a single node in the interior, resulting in smaller amplitudes and shorter periods (roughly 70% of the fundamental period for a star of the same brightness).

Because the period ratio P_1/P_0 is highly sensitive to stellar mass and chemical composition, some Cepheids (known as double-mode or beat Cepheids) pulsate in both modes simultaneously, offering stringent constraints on stellar structure models [1].

In modern catalogues, the standard notation used is:

- **DCEP** — classical Cepheid, fundamental mode
- **DCEPa** (or **DCEP_1O**) — classical Cepheid, first overtone

1.6 Chemical Abundances and Galactic Science

Beyond their utility as distance indicators, classical Cepheids are powerful tracers of the chemical composition of the present-day Galactic disk. Because they are relatively young (with ages ranging from tens to hundreds of Myr), they reflect the current metallicity of the interstellar medium at their respective Galactic radii. This makes them ideal probes of the radial metallicity gradient—the systematic decrease in metal abundance from the inner Galaxy out to the outer disk.

Key findings from Cepheid abundance surveys include:

- The Galactic disk exhibits a radial iron gradient of approximately -0.06 dex/kpc (declining outwards), although the slope flattens beyond a Galactocentric radius of ~ 10 kpc [12].
- Alpha-element abundance ratios ($[\alpha/\text{Fe}]$) are approximately solar in Galactic Cepheids, confirming they formed from well-mixed disk material [12, 36].
- Individual elemental ratios (e.g., Ba, La, Y, Eu) trace the contributions of s -process and r -process nucleosynthesis as a function of Galactic radius and age, providing vital constraints on chemical evolution models [36].

The C-MetaLL survey (Chemical abundances of Metal-poor, Long-period, Large-amplitude Cepheids; [28]) specifically targets DCEPs located towards the metal-poor outer disk and halo, extending chemical mapping to Galactocentric distances beyond 15 kpc where very few abundance measurements previously existed [36].

1.7 Project Objectives

The primary objectives of this project are to:

- Analyse the observed high-resolution spectra of the target star.
- Measure the equivalent widths of neutral and ionised iron lines using the automated ARES code with customised continuum fitting.
- Determine fundamental atmospheric parameters (T_{eff} , $\log g$, and ξ_t) under LTE assumptions using 1D Kurucz ATLAS9 model atmospheres.
- Implement an automated Python code to iteratively optimise parameter convergence using excitation and ionisation equilibrium conditions.
- Develop a synthetic (best-fit) spectrum using the optimised parameters.

- Model macroturbulent line broadening (V_{broad}) by synthesising stellar profiles using the MOOG `synth` driver.

Chapter 2

Theoretical Background of Spectroscopy

2.1 Fundamentals of Stellar Spectroscopy

Stellar spectroscopy is fundamentally based on the fact that electromagnetic emission or absorption lines emanating from a star are rich in physical information regarding its outer atmosphere. The physical processes that occur when light interacts with atoms and ions during its journey through the various plasma layers involve bound-bound transitions, resulting in specific absorption lines superimposed on a continuous spectrum. The wavelengths, depths, and shapes of these absorption lines depend entirely on the physical conditions of the absorbing gas, including its temperature, pressure, composition, and dynamics.

The underlying theory was developed in the early twentieth century through advances in quantum mechanics and thermodynamics. According to Niels Bohr's quantisation postulate, electrons can only occupy definite energy levels and will radiate or absorb light when transitioning between them. In fact, the wavelength of the emission or absorption can be calculated directly from the energy difference between the upper level u and the lower level l :

$$\lambda = \frac{hc}{E_u - E_l} \quad (2.1)$$

where h is Planck's constant, c is the speed of light, and the energy levels are characteristic of each atomic species. This quantum structure ultimately makes the stellar spectrum a unique chemical fingerprint [30, 13].

2.2 Radiative Transfer in Stellar Atmospheres

The transport of radiation through a star's atmosphere is described by the equation of radiative transfer. For stars that are neither extremely hot nor cold (such as classical Cepheids), this equation in a plane-parallel geometry can be written as:

$$\mu \frac{dI_\nu}{d\tau_\nu} = I_\nu - S_\nu \quad (2.2)$$

where:

- I_ν is the specific intensity (brightness) at frequency ν ,
- τ_ν is the monochromatic optical depth,
- $\mu = \cos \theta$ is the direction cosine of the ray,
- S_ν is the source function.

If we assume Local Thermodynamic Equilibrium (LTE), the source function simplifies to the Planck function:

$$S_\nu = B_\nu(T) \quad (2.3)$$

which depends solely on the local kinetic temperature T .

The formal solution for the emergent radiation (I_ν leaving the surface at $\tau_\nu = 0$) is:

$$I_\nu(0, \mu) = \int_0^\infty S_\nu(\tau_\nu) e^{-\tau_\nu/\mu} \frac{d\tau_\nu}{\mu} \quad (2.4)$$

Under the Eddington–Barbier approximation, this integral simplifies to:

$$I_\nu(0, \mu) \approx S_\nu(\tau_\nu = \mu) \quad (2.5)$$

This relation tells us that the light we observe at a given wavelength or direction originates approximately from the layer where the optical depth is $\tau_\nu = \mu$ [13, 25] (or $\tau_\nu = 1$ for vertical rays where $\mu = 1$).

This has critical implications for how spectral lines are formed:

- **In absorption lines:** The opacity is high, so we reach $\tau_\nu = 1$ at shallow, cooler outer layers of the atmosphere, resulting in a weaker emergent intensity (the absorption core).
- **In the continuum:** The opacity is lower, allowing us to see deeper into the hotter regions of the photosphere, producing a stronger emergent intensity.

Consequently, the contrast between the line core and the surrounding continuum

provides direct clues about the temperature gradient and the opacity structure within the stellar atmosphere.

2.3 Local Thermodynamic Equilibrium (LTE) and its Applicability

A standard working assumption in classical stellar abundance analysis is Local Thermodynamic Equilibrium (LTE). In this framework, the population levels of atomic species are assumed to follow the Boltzmann distribution, the ionisation states are described by the Saha equation, and the local source function is strictly thermal ($S_\nu = B_\nu(T)$) at the local kinetic temperature T . This assumption is physically valid when collisional interactions between particles dominate over radiative transitions in regulating the excitation and ionisation states.

The population ratio of an excited state i relative to the ground state 1 is given by the Boltzmann equation:

$$\frac{N_i}{N_1} = \frac{g_i}{g_1} e^{-(E_i - E_1)/kT} \quad (2.6)$$

where g represents the statistical weight of the respective energy level, and k is the Boltzmann constant.

Similarly, the ionisation balance between successive ionisation stages is governed by the Saha equation:

$$\frac{N_{r+1}}{N_r} = \frac{2kT}{P_e} \frac{U_{r+1}(T)}{U_r(T)} \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-\chi_r/kT} \quad (2.7)$$

where P_e is the electron pressure, $U_r(T)$ and $U_{r+1}(T)$ are the partition functions of the successive stages, m_e is the electron mass, and χ_r is the ionisation potential of the lower stage r [25, 2].

While LTE simplifies abundance determinations, physical departures (non-LTE or NLTE effects) are expected in the outer, optically thin layers of the stellar atmosphere where the radiation field decouples from the local thermal pool. Nonetheless, for the neutral and singly ionised iron lines in the optical range (4000–7000 Å) routinely used in Cepheid analyses, departures from LTE are typically minor. These NLTE corrections are generally smaller than or comparable to other systematic modelling uncertainties [35, 3]. While a full NLTE analysis is beyond the scope of this project, literature results confirm that LTE-derived iron abundances for Cepheids remain internally consistent to within 0.05–0.10 dex under these physical conditions.

2.4 Line Formation

Multiple simultaneous broadening mechanisms shape an absorption line profile. For the majority of spectral lines, natural broadening—which results from the excited state’s finite lifetime—produces a Lorentzian profile with a minimal width. Thermal (Doppler) broadening, caused by the Maxwell–Boltzmann distribution of atomic velocities along the line of sight, yields a Gaussian core:

$$\Delta\lambda_D = \frac{\lambda}{c} \sqrt{\frac{2kT}{m_a} + \xi^2} \quad (2.8)$$

where m_a is the mass of the absorbing atom, T is the local kinetic temperature, k is the Boltzmann constant, c is the speed of light, and ξ is the microturbulent velocity parameter, which accounts for small-scale, unresolved convective motions within the photosphere [13].

At higher column densities, pressure broadening—especially the van der Waals interaction between an absorber and surrounding neutral hydrogen—contributes Lorentzian wings. The combined Voigt profile is the convolution of the Gaussian and Lorentzian components and is the profile shape routinely fitted in high-resolution work.

2.5 Equivalent Width

The equivalent width (EW) is the primary measurable quantity obtained from each absorption line used in this study. It is defined as the width of a rectangular notch, dropping from the continuum to zero intensity, that removes the same amount of flux as the true absorption line:

$$W_\lambda = \int \left(1 - \frac{F_\lambda}{F_c}\right) d\lambda \quad (2.9)$$

where F_λ is the flux inside the line and F_c is the adjacent continuum flux.

The relationship between the equivalent width of a line and the abundance of the absorbing species is described by the curve of growth. For weak spectral lines (the linear regime), W_λ scales linearly with the abundance and the oscillator strength of the transition, making them excellent abundance indicators. As the line becomes stronger, its core saturates (the flat regime), and W_λ becomes nearly independent of abundance, instead depending heavily on the microturbulent velocity ξ .

For very strong lines, the wings dominate (the damping or square-root regime), where W_λ grows with the square root of the abundance and depends on the damping constant. In Cepheid analyses, we prioritise using weak-to-intermediate iron lines that lie mostly in the linear and transition regimes to minimise systematic errors from microturbulence

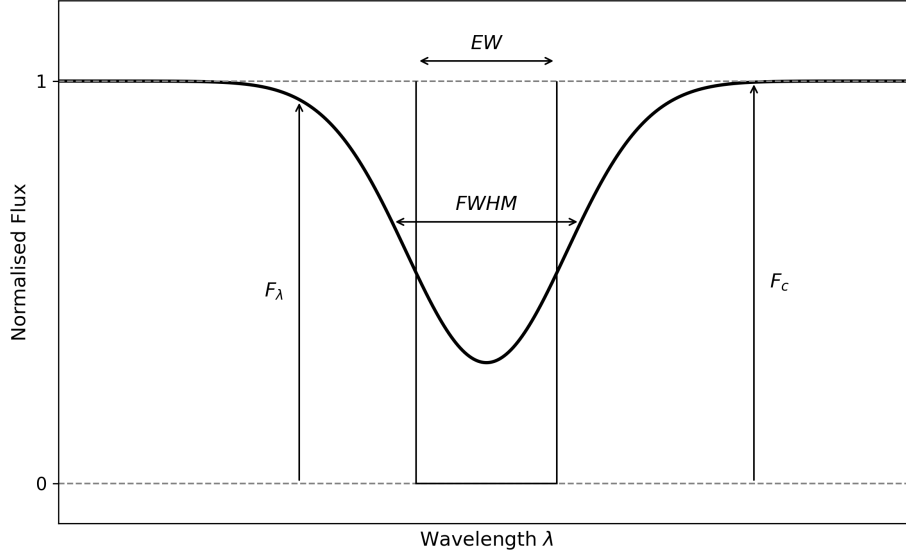


Figure 2.1: The equivalent width of an absorption line is defined as the width of a rectangle that has an area equal to the line, as illustrated in grey. Adapted from Figure 9.18 of [6].

and damping parameters.

2.6 Curve of Growth

The relationship between the equivalent width (W_λ) of a spectral line and the abundance of the absorbing element (expressed through the column density) is described by the curve of growth (COG). To establish its physical basis, we can model the line profile in a simplified two-component atmosphere (consisting of a continuum-emitting photosphere and an absorbing layer of thickness h). The rectified line profile is given by:

$$r_\lambda = \frac{I_\lambda}{I_c} = e^{-\kappa_l(\lambda)h} \quad (2.10)$$

where $\kappa_l(\lambda)$ is the line absorption coefficient. Integrating over the line profile yields the equivalent width:

$$W_\lambda = \int_0^\infty (1 - e^{-\kappa_l(\lambda)h}) d\lambda \quad (2.11)$$

Depending on the optical depth at the line centre ($\tau_0 \propto Nf\lambda$), the COG is divided into three distinct physical regimes [13, 30]:

- **Linear Regime** ($\tau_0 \ll 1$): For weak, optically thin lines, we can approximate $1 - e^{-\kappa_l h} \approx \kappa_l h$. The equivalent width grows in direct proportion to the abundance ($W_\lambda \propto A$):

$$W_\lambda \approx h \int_0^\infty \kappa_l(\lambda) d\lambda \propto N h f \lambda_0^2 \quad (2.12)$$

where N is the number density of absorbers and f is the oscillator strength. Here,

the profile remains Gaussian and unsaturated.

- **Flat Regime** ($10 \lesssim \tau_0 \lesssim 10^3$): As the abundance increases, the line core saturates. Once almost all photons at the line centre are absorbed, the equivalent width increases very slowly ($W_\lambda \propto \sqrt{\ln A}$). In this regime, W_λ is highly sensitive to the microturbulent velocity ξ , which broadens the Doppler core and delays saturation.
- **Damping Regime** ($\tau_0 \gg 10^3$): For very strong lines, the optical depth in the Lorentzian wings becomes significant. The equivalent width begins to grow again as the square root of the column density ($W_\lambda \propto \sqrt{A}$) and is governed by the damping constant Γ .

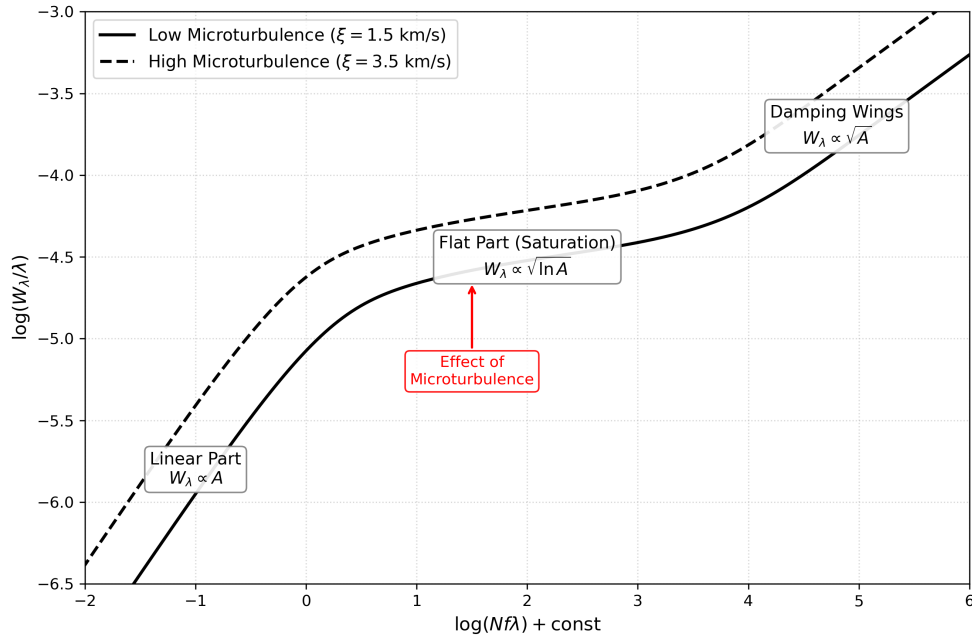


Figure 2.2: Theoretical Curve of Growth calculated for a Voigt profile showing the linear, flat, and damping regimes. The solid curve shows a low-microturbulence model ($\xi = 1.5$ km/s), while the dashed curve demonstrates how an increased microturbulent velocity ($\xi = 3.5$ km/s) delays saturation and shifts the flat plateau upward.

The shape and position of the COG are also sensitive to other atmospheric parameters:

- **Temperature:** Temperature shifts the COG shape horizontally due to changes in the population of the lower excitation states, described by the Boltzmann distribution:

$$\log \left(\frac{W_\lambda}{\lambda} \right) = \log A + \log(gf\lambda) - \chi\theta_{\text{exc}} + \text{const} \quad (2.13)$$

where $\theta_{\text{exc}} = 5040/T_{\text{exc}}$.

- **Gravity:** Gravity affects the ionisation state of the gas via the Saha equation. This pressure effect is primarily important for ionised species (Fe II), while remaining

largely cancelled out by opacity shifts for neutral lines (Fe I).

The iron lines selected for this analysis span excitation potentials from approximately 0.8 to 6.0 eV and equivalent widths from roughly 12 to 200 mÅ. This range places them predominantly in the linear-to-flat transition region of the COG (see Figure 2.2). This positioning is ideal because it allows us to constrain the microturbulent velocity and iron abundance simultaneously. Lines with $W_\lambda > 200$ mÅ are excluded, as they reside in the damping regime where abundance sensitivity is low and uncertainties in pressure-broadening parameters are high.

2.7 Model Atmospheres

A model atmosphere serves as a grid of physical parameters, specifying temperature, gas and electron pressure, and opacity as a function of depth within a simplified, static, one-dimensional stellar photosphere. The depth in these models is typically parameterised using either the Rosseland mean optical depth τ_R or the mass column density. In stellar spectroscopy, these models are indispensable; modelling or interpreting any spectral line requires a detailed map of the temperature and pressure structure across the entire region of the atmosphere where the line is formed.

For the spectroscopic analysis in this work, we adopt the standard ATLAS9 grids of plane-parallel, line-blanketed model atmospheres in LTE, originally developed by Kurucz [19] and subsequently updated by Castelli & Kurucz [7]. The calculations for these models enforce hydrostatic equilibrium, strict conservation of radiative and convective flux, and utilise mixing-length theory to model convective energy transport. Each model atmosphere is defined by four fundamental parameters: the effective temperature (T_{eff}), the surface gravity ($\log g$), the microturbulent velocity (ξ), and the global metallicity ($[M/H]$). Specifically, we employ the solar-scaled abundance grid (AODFNEW) without α -element enhancement, which is the standard choice in classical Cepheid studies (e.g., [23, 12, 8]).

Chapter 3

Instrumentation, Observations, and Data Reduction

3.1 The Canada-France-Hawaii Telescope

Data for the target star were obtained using the Canada-France-Hawaii Telescope (CFHT), a 3.6-metre optical/infrared telescope located at the summit of Mauna Kea, Hawaii, at an altitude of 4,204 metres above sea level. The telescope has been in continuous scientific operation since 1979 and is operated as an international partnership between the National Research Council of Canada, the Centre National de la Recherche Scientifique of France, and the University of Hawaii.



Figure 3.1: The Canada-France-Hawaii Telescope (CFHT) dome at dusk on the summit of Mauna Kea, Hawaii. (Image credit: Jean-Charles Cuillandre / CFHT).

The high altitude and stable atmospheric conditions at Mauna Kea produce exceptionally low atmospheric water vapour content and median seeing typically below 0.7 arcseconds. These properties make the site particularly well suited to optical spectroscopy, where sky background and atmospheric dispersion must both be minimised to preserve spectral quality. The telescope employs an alt-azimuth mount and feeds instruments at the Cassegrain focus, where the spectrograph used in this project is permanently installed.

3.2 The ESPaDOnS Spectrograph

3.2.1 Overview

The ESPaDOnS (Echelle SpectroPolarimetric Device for the Observation of Stars) instrument at the CFHT is a fibre-fed, cross-dispersed échelle spectrograph and spectropolarimeter. The spectrograph unit itself is permanently mounted in a thermally controlled, vibration-isolated Coudé room on the third floor of the telescope dome (see Figure 3.2b), while the polarimeter unit is mounted at the Cassegrain focus of the telescope (see Figure 3.2a). Optical fibres (30 metres in length) connect the Cassegrain unit to the spectrograph bench. The instrument was designed and built by Jean-François Donati (IRAP, Toulouse) and commissioned in 2004 [9]. Since its commissioning, it has been one of the most heavily subscribed instruments at the telescope, used widely for studies of stellar magnetic fields, chemical abundances, and pulsating variables.



(a) Cassegrain polarimeter unit.



(b) Spectrograph optical bench.

Figure 3.2: Key hardware components of the ESPaDOnS instrument at the CFHT (Image credits: CFHT).

ESPaDOnS operates in two primary observing modes: **spectropolarimetric mode** and **spectroscopic mode**. In the spectropolarimetric mode, circularly or linearly polarised light is separated by a set of Fresnel rhombs and a Wollaston prism before entering the spectrograph fibres. In the spectroscopic mode, the instrument can be configured in either “Star-only” mode (which uses a 6-slice image slicer to deliver maximum spectral resolution, $R \approx 80,000$) or “Star+Sky” mode (which uses a 3-slice image slicer for higher throughput, yielding a resolving power of $R \approx 68,000$). The spectra analysed in this thesis were obtained in the “Star+Sky” spectroscopic mode to maximise throughput while maintaining a high resolving power.

3.2.2 Optical Design

The instrument consists of two main components: a Cassegrain polarimeter unit mounted directly at the Cassegrain focus of the telescope, and a bench-mounted spectrograph located in a thermally controlled, vibration-isolated Coudé room on the third floor of the dome. The two units are connected by a fibre-optic cable approximately 30 metres in length. The spectrograph itself utilises a dual-pupil, quasi-Littrow configuration that features a fully dioptric camera to avoid central obscuration.

At the Cassegrain unit, the stellar light is collected through a 1.6-arcsecond on-sky aperture and fed into the spectrograph fibre. In the “Star+Sky” spectroscopic mode, a second fibre simultaneously samples the sky background through a 2.2-arcsecond aperture located 7.9 arcseconds away, enabling accurate sky subtraction for faint targets.

In the spectrograph, the light from the fibre is collimated and dispersed by an échelle grating (79 lines per mm), then cross-dispersed by a series of prisms that separate the overlapping diffraction orders in the perpendicular direction. The resulting two-dimensional pattern of spectral orders is imaged onto the CCD detector. The cross-dispersion by prisms (rather than a grating) provides a near-constant spectral format across the full wavelength range, simplifying the wavelength calibration.

3.2.3 Detector and Wavelength Coverage

For the observations analysed in this work, the detector used was *Olapa*, a deep-depletion e2v 2048×4096 pixel CCD with $15\ \mu\text{m}$ pixels, which replaced the original EEV1 detector in 2011. The *Olapa* detector delivers significantly higher quantum efficiency in the red and exhibits minimal fringing compared to its predecessor. The spectral format produces exactly 40 échelle orders (spanning from order #61 in the blue to order #22 in the red), covering the full wavelength range from 370 nm to 1050 nm in a single exposure. This continuous coverage—from the near-ultraviolet through the red and into the near-infrared—is one of the primary advantages of ESPaDOnS for stellar spectroscopy, as it allows iron lines, hydrogen Balmer lines, calcium lines, and other diagnostic features to be observed simultaneously.

The signal-to-noise ratio (SNR) varies substantially across the spectral range, being lowest in the blue (where the CCD quantum efficiency and stellar flux are both low) and peaking in the red near 700–750 nm, where the silicon-based detector responds most efficiently.

3.3 Resolving Power

The spectroscopic (Star-only) mode of ESPaDOnS delivers a resolving power of $R \approx 80,000$, corresponding to a velocity resolution of approximately 3.7 km/s per resolution element. This is sufficient to resolve individual iron lines in the spectra of cool supergiants, measure radial velocities to sub-km/s accuracy after cross-correlation, and detect profile broadening due to macroturbulence in classical Cepheids (which typically ranges between 8–15 km/s). The high resolving power also reduces the risk of line blending in crowded spectral regions, which would otherwise introduce biases into equivalent width measurements.

Table 3.1: Summary of ESPaDOnS instrument parameters in the Star-only (high-resolution) spectroscopic configuration.

Parameter	Value
Telescope aperture	3.6 m
Fibre diameter on sky (aperture)	1.6 arcsec
Wavelength coverage	370–1050 nm
Number of échelle orders	≈ 40
Resolving power (R)	$\approx 80,000$
Velocity resolution	≈ 3.7 km/s
Image slicer configuration	6 slices
Detector	e2v deep-depletion CCD (Olapa)
Detector format	2048×4608 pixels
Pixel size	$13.5 \mu\text{m}$

3.4 Target Star: SV Persei

3.4.1 General Properties

The target of this project is SV Persei (SV Per), a classical Cepheid variable star located in the Perseus constellation. SV Per is a fundamental-mode classical Cepheid (DCEP). As a Galactic disk Cepheid, it falls within the Cepheid Metallicity in the Leavitt Law (C-MetaLL) survey sample [37], which targets classical Cepheids across a wide range of Galactic radii to characterise the radial metallicity gradient of the present-day Milky Way disk.

3.4.2 Pulsation Phase

A key observational challenge with Cepheid spectroscopy is that the stellar atmosphere is never in a static state. The photosphere expands and contracts over each pulsation cycle, causing the effective temperature, surface gravity, radial velocity, and line-broadening

parameters to vary continuously with phase. Single-epoch observations, therefore, sample only one point in this cycle.

3.4.3 Data Reduction: The Libre-ESPRIT Pipeline

The spectroscopic data analysed in this work were obtained as pre-processed, fully reduced, wavelength-calibrated, and continuum-normalised spectra directly from the CFHT data archive. The raw CCD frames were processed automatically upon readout by the CFHT automated pipeline *Upena* (version 1.1), which utilises the *Libre-ESPRIT* software package [10] for data reduction. Consequently, no manual raw frame reduction was required. The standard reduction sequence performed automatically by the pipeline to produce the analysed data products is described below.

Bias Subtraction

Each science frame is preceded by a set of bias exposures (zero-second dark frames taken at the start of each night) that characterise the baseline electronic offset of the two CCD amplifiers. The median bias level, measured in the overscan regions at the edges of each frame, is subtracted row by row from all subsequent frames.

Flat-Field Correction

Pixel-to-pixel variations in detector sensitivity are corrected using halogen lamp flat-field frames, taken at the beginning of each observing run with the calibration unit built into the Cassegrain module. The flat fields are also used to trace the positions of the échelle orders on the CCD and to construct the extraction aperture for each order. Because the instrument is fibre-fed and thermally stabilised, the spectral format is highly stable from night to night, and a single set of flat-field frames is generally sufficient for the full observing run.

Wavelength Calibration

Wavelength calibration is performed using Thorium-Argon (ThAr) hollow-cathode lamp exposures. The ThAr spectrum consists of hundreds of well-characterised emission lines distributed across the full wavelength range of the spectrograph, providing a dense reference grid from which a two-dimensional wavelength solution is derived as a polynomial function of pixel position on the CCD. The typical RMS residual of the wavelength solution is below 30 m/s.

ThAr calibration frames are taken immediately before and after each science observation to account for any small instrumental drift during the night, though the thermal stability of ESPaDOnS keeps such drifts to well below one pixel.

Optimal Spectral Extraction

The one-dimensional spectrum in each échelle order is extracted using the optimal extraction algorithm of Horne (1986) [15], as implemented in Libre-ESPRIT. Optimal extraction weights each pixel in the spatial profile of the order by its signal-to-noise ratio, suppressing the contribution from the noisy wings of the profile while maximising the statistical weight of the well-illuminated core. This produces extracted spectra with an effectively better SNR compared to simple aperture summation, particularly for faint sources or in blue orders where the photon count is low.

The sky background is simultaneously extracted from a second fibre (where applicable, such as in standard dual-fibre configurations) and subtracted from the stellar spectrum. For observations of bright Cepheid variables like SV Persei, the sky background contribution is negligible, but the automated pipeline includes this subtraction step to ensure that any minor scattered light or background contamination is removed from the final data products.

Continuum Normalisation

After extraction, each échelle order is normalised to a unit continuum level. The pipeline fits a low-order polynomial to the continuum regions of each order, identified automatically by iterative sigma-clipping to exclude absorption lines. The resulting normalised spectra have a continuum level of approximately 1.0 across each order.

3.5 Extracted Spectra: Properties and Quality

Echelle Order Structure

The reduced ESPaDOnS spectrum of SV Persei consists of 40 individual échelle orders, each covering a wavelength range of approximately 20–30 nm, with small gaps between consecutive orders in the red portion of the spectrum where the orders diverge. Each order is stored as a separate one-dimensional array of wavelength and normalised flux values, enabling targeted analysis of any desired spectral region.

For the iron-line abundance and line-broadening analysis carried out in this work, spectral windows from approximately 400 nm to 900 nm were used, corresponding to échelle orders 25 to 56. The spectral region below 400 nm was excluded from the analysis as it is dominated by noise (with an SNR below ≈ 20). This range was chosen because it contains the largest number of well-isolated Fe I and Fe II lines from our linelist, and because the SNR in this region is consistently above ≈ 30 —the minimum threshold for reliable measurements.

Signal-to-Noise Ratio as a Function of Wavelength

The per-pixel SNR in each order is recorded in the FITS file header of every reduced ESPaDOnS spectrum. Two SNR values are reported per order, corresponding to the signal-to-noise ratio per spectral pixel in the extracted one-dimensional spectrum, and per CCD pixel in the raw two-dimensional frame. Table 3.2 lists a representative subset of these values for the epoch analysed in detail in this thesis (HJD 2459838.14255).

Table 3.2: Signal-to-noise ratio (SNR) values per spectral pixel and per CCD pixel across different échelle orders for the SV Persei spectrum.

Order	Central Wavelength (nm)	SNR (per spectral pixel)	SNR (per CCD pixel)
56	404	28	33
52	435	41	48
48	472	65	77
44	515	84	100
40	566	106	126
36	629	121	144
32	708	145	173
28	809	157	188
25	906	158	189

As shown in Table 3.2, the SNR increases significantly towards longer wavelengths. This behaviour is primarily a consequence of two physical factors: the energy distribution of cool stars (such as *F*- and *G*-type Cepheids), which peaks in the optical/near-infrared, and the quantum efficiency of the deep-depletion Olapa CCD detector, which is optimised for red wavelengths. The high SNR in the red orders (SNR > 100) makes them particularly suited for precise equivalent width and line profile analysis.

Chapter 4

Methodology

4.1 Overview

The spectroscopic analysis of SV Persei carried out in this work follows a four-phase semi automated pipeline, which starts from the assembly of observational data and theoretical models through to the measurement of atmospheric parameters, iron abundances, and developing a synthetic spectrum for the optimized parameters. The entire work is implemented in Python 3 along with the FORTRAN-based MOOG spectral analysis code through the pymoog wrapper.

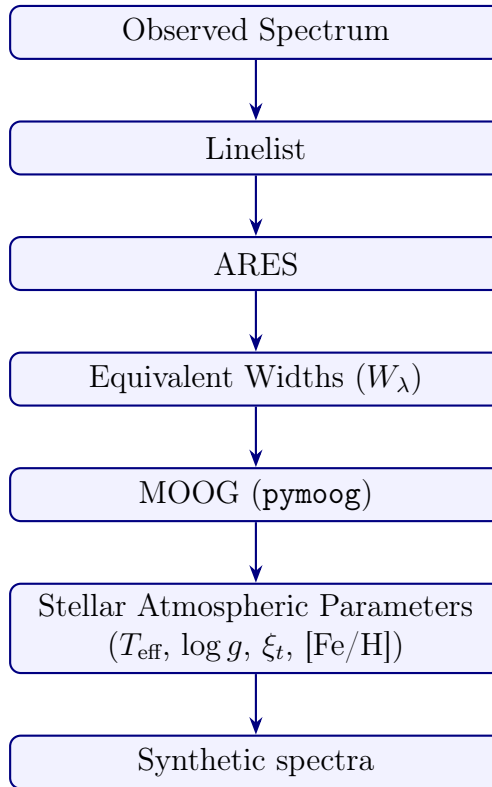


Figure 4.1: Flowchart summarizing the spectroscopic analysis workflow for SV Persei.

4.2 Phase 1: Inputs

4.2.1 Overview of Observed spectrum

The starting point for the entire analysis is the reduced, radial-velocity corrected, one-dimensional ESPaDOnS spectrum of SV Persei of epoch(HJD 2459838.14255), described in Chapter 3. The spectrum is stored as a two-column plain-text file containing rest-frame wavelength ($\text{\AA}$) and continuum-normalized flux.

4.2.2 Radial Velocity Estimation and Rest-Frame Correction

In high-resolution stellar spectroscopy, observed absorption features are systematically Doppler-shifted due to the relative line-of-sight motion between the star and the observer. For a pulsating classical Cepheid like SV Persei, this macroscopic radial velocity (V_r) is highly dynamic; it varies significantly across the pulsation cycle as the stellar photosphere continuously expands and contracts. Consequently, before extracting equivalent widths or performing theoretical spectral synthesis, the observed spectrum must be precisely shifted to the stellar rest frame.

Because the radial velocities of Cepheids (typically tens of km/s) are vastly smaller than the speed of light ($V_r \ll c$), the shift can be accurately described by the non-relativistic Doppler equation:

$$V_r = c \left(\frac{\lambda_{\text{obs}} - \lambda_{\text{lab}}}{\lambda_{\text{lab}}} \right) \quad (4.1)$$

Within our automated Python pipeline, the RV estimation is handled via template cross-correlation prior to the primary abundance analysis. Instead of measuring individual line cores, the normalised observed spectrum is cross-correlated against a theoretical synthetic template generated by the MOOG radiative transfer code. This approach utilises the combined signal of all absorption features within the selected spectral window, making it highly robust against noise and localized blending.

The peak of the resulting Cross-Correlation Function (CCF) indicates the global radial velocity shift. To achieve sub-pixel precision, our algorithm fits a Gaussian profile to the core of the CCF peak. The centroid of this Gaussian yields the measured radial velocity. Furthermore, the formal 1σ uncertainty of this measurement is calculated by combining the covariance error of the Gaussian fit and the photon-noise-limited uncertainty (derived from the spectral signal-to-noise ratio following the methodology of Zucker, 2003) in quadrature.

It is important to note that the raw observed velocity includes the kinematic motion of the Earth (its rotation and its orbit around the barycentre of the Solar System). However, the spectra processed by the CFHT *Libre-ESPRIT* pipeline are already

corrected for this effect, meaning the wavelength grid provided in the FITS files is barycentric. Therefore, the V_r calculated by our pipeline represents the true topocentric radial velocity of the star relative to the Solar System barycentre.

Once V_r is quantified, the entire wavelength array of the ESPaDOnS spectrum is mathematically shifted to the rest frame ($V_r = 0$). This rest-frame correction is a critical prerequisite for the subsequent steps of the methodology, as it ensures the automated Gaussian fitting routines in **ARES** can accurately cross-reference the observed absorption features with the static wavelengths provided in the Genovali et al. (2014) [12] reference line list.

4.2.3 Genovali Line-list

The reference set of spectral lines used throughout this work is the Fe I and Fe II line-list published by Genovali et al. (2014) [12], which is a standard line-list for Galactic Cepheid chemical abundance studies and is the same list adopted by the C-MetaLL survey (Trentin et al. 2024) [37]. The list was constructed from the compilations of Romaniello et al. (2008) [29] and Luck & Heiter (2007) [22], with oscillator strengths ($\log gf$) drawn primarily from the laboratory measurements of Fuhr & Wiese (2006) [11] and the NIST Atomic Spectra Database [18]. It contains approximately 283 transitions distributed throughout the wavelength range of 4800–9900 Å, comprising both neutral iron (Fe I, species code 26.0) and single-ionized iron (Fe II, species code 26.1).

Table 4.1: Sample lines from the curated Genovali et al. (2014) iron linelist.

Wavelength (Å)	Species	Excitation Potential (eV)	$\log gf$ (dex)
4892.87	Fe I (26.0)	4.22	−1.29
4893.82	Fe II (26.1)	2.82	−4.27
4917.23	Fe I (26.0)	4.19	−1.18
4923.93	Fe II (26.1)	2.89	−1.50
4924.77	Fe I (26.0)	2.28	−2.24
4932.08	Fe I (26.0)	4.65	−1.49
4950.10	Fe I (26.0)	3.42	−1.67
4973.11	Fe I (26.0)	3.96	−0.95
4993.35	Fe II (26.1)	2.81	−3.68
4994.13	Fe I (26.0)	0.91	−3.08

4.2.4 Model Atmosphere grid: Kurucz ATLAS9

The framework for interpreting the equivalent widths requires some model of the stellar photosphere, which is the description of how the dependencies of temperature, pressure, gas density, and electron density vary with depth. In this work, one-dimensional plane-parallel hydrostatic atmospheres are used which were constructed within the framework

of the Kurucz ATLAS9 grid [7]. These models are computed under the assumption of Local Thermodynamic Equilibrium (LTE).

The ATLAS9 grid contains the model atmospheres for effective temperatures between 3500 and 50 000 K in steps of 250 K, for surface gravity $\log g$ between 0.0 and 5.0 in steps of 0.5 dex, and for metallicities $[M/H]$ from -2.5 to $+0.5$ in steps of 0.5 dex. The atmospheres in the case when trial parameters do not coincide exactly with any point of the grid (which inevitably happens in the optimization process) are built by means of trilinear interpolation in parameters $\log(T_{\text{eff}})$, $\log g$, and $[Fe/H]$.

4.3 Phase 2: Equivalent Width Measurement

4.3.1 The Role of Equivalent Widths

As discussed in section 2.5, the equivalent width (W_λ) of a spectral absorption line is defined as the width of a rectangular notch in the continuum that would remove the same total flux as the actual line profile:

$$W_\lambda = \int \left(1 - \frac{F_\lambda}{F_{\text{cont}}} \right) d\lambda \quad (4.2)$$

where F_λ is the observed flux and F_{cont} is the local continuum flux level. Expressed in milli-Ångströms (mÅ), W_λ is a direct empirical measure of the total absorption strength of the line, integrated over its full profile regardless of the broadening mechanisms that shape it.

For optimization of atmospheric parameters (Phase 3), I have restricted this analysis to lines with equivalent widths satisfying $W_\lambda < 200$ mÅ. This criterion ensures that the selected transitions lie primarily on the linear and lower parts of the curve of growth, where W_λ remains directly proportional and highly sensitive to the chemical abundance of the absorbing species. Saturated lines exceeding this threshold were excluded from the optimization process, as their equivalent widths are dominated by microturbulent velocity and damping rather than the elemental abundance, which would introduce systematic biases into the final stellar parameters.

4.3.2 Overview of ARES

ARES (Automatic Routine for line Equivalent widths in stellar Spectra) is an automated line-measurement code written in C by Sousa et al. [32, 33]. Given a normalized one-dimensional spectrum and a reference line-list, ARES identifies the spectral region surrounding each listed wavelength, fits a Gaussian profile to the absorption feature, and returns the equivalent width as the area under the fitted Gaussian.

The Gaussian fitting model is appropriate for isolated, unblended lines in the EW range 50–150 mÅ, where the profile shape is governed primarily by Doppler broadening (thermal and microturbulent). For stronger lines approaching the flat part of the curve of growth, Voigt profiles would better capture the extended damping wings; however, as described in Section 4.3.5, lines in this work are accepted over the range 10–200 mÅ, with the upper bound set to retain a broader sample of lines while relying on the sigma-clipping step to remove any individual measurements whose profiles are too strongly non-Gaussian for a reliable Gaussian fit. The primary advantage of **ARES** over manual measurement is strict reproducibility: given fixed input parameters, the output is identical for every run, eliminating the analyst-to-analyst variability that characterizes manual EW estimation.

4.3.3 ARES Configuration

ARES operation is controlled by several key parameters in its input file called `mine.opt`. These parameters and the values used for this work is listed in the following table.

Table 4.2: ARES configuration parameters.

Parameter	Value	Role
<code>lambdai</code>	4000 Å	Start of analysis range
<code>lambdaf</code>	9900 Å	End of analysis range
<code>smoothder</code>	4	Smoothing for derivative-based line detection
<code>space</code>	2.0 Å	Half-width of local continuum window
<code>rejt</code>	0.995	Continuum rejection percentile
<code>lineresol</code>	0.1 Å	Minimum separation for two resolved features
<code>miniline</code>	10.0 mÅ	Minimum EW to report

The most important parameter is `rejt`, which controls the percentile level used internally by **ARES** to distinguish the continuum from absorption features within the `space` window. For a spectrum with SNR ≈ 100 –130, the theoretically optimal value is $\text{rejt} \approx 1 - 1/\text{SNR} \approx 0.993$ –0.997, following the calibration procedure of Sousa et al. (2008) [34]. The value `rejt` = 0.995 was adopted as a suitable value for the SNR range of the SV Persei data.

4.3.4 Filtering of ARES Output

The raw **ARES** output contains all detected features satisfying the `miniline` threshold, regardless of quality. Before any lines are passed to the MOOG analysis, a set of quality filters is applied.

EW range: 10–200 mÅ. The lower bound of 10 mÅ is set to exclude features at or below the noise floor: at the per-pixel SNR of 120–130 achieved in this dataset,

the measurement uncertainty on a Gaussian fit approaches several mÅ, making lines weaker than ≈ 10 mÅ unreliable. The upper bound of 200 mÅ is deliberately inclusive compared to the more conservative 150 mÅ limit used in some studies, reflecting the lower surface gravity of SV Persei as a supergiant ($\log g \approx 0.44$) [36] relative to dwarf or sub-giant targets. At low gravity, the reduced gas pressure suppresses pressure-broadened damping wings even in moderately strong lines, keeping their profiles closer to Gaussian up to larger EW values. Lines exceeding 200 mÅ are excluded because at this strength, even in low-gravity atmospheres, the curve of growth is visibly flattening and the EW is no longer a reliable linear abundance indicator. Any residual outliers within the accepted range that deviate from a Gaussian profile are caught and removed by the subsequent MAD sigma-clipping step.

Excitation potential: $EP > 2.5$ eV. Low-EP lines form in the outermost, coolest layers of the atmosphere and are more susceptible to model-atmosphere errors in the temperature gradient. Restricting the analysis to higher-EP lines improves the robustness of the excitation equilibrium diagnostic.

MAD sigma-clipping: 3σ rejection. After a first-pass abundance calculation, lines whose derived abundance deviates from the median by more than 3 times the median absolute deviation (MAD) are removed. This step is applied once, after which the atmospheric parameter fitting is repeated on the cleaned sample.

The output of Phase 2 is a quality-filtered 7-column line-list containing the wavelength, species code, excitation potential, $\log gf$, two damping constants (zero), and measured EW for each accepted line. This file serves as the primary input to Phase 3 and is formatted directly in the MOOG input style, so that no intermediate reformatting is required.

4.4 Phase 3: Atmospheric Parameter Determination

4.4.1 The abfind Driver

The core of Phase 3 is the MOOG abfind driver, accessed through pymoog. Given a model atmosphere, a linelist with measured EW values, and a set of atomic parameters, abfind solves for the elemental abundance that causes the theoretical equivalent width computed by numerically integrating the emergent line profile over the model atmosphere to match the observed W_λ for each line individually. The result is a per-line abundance $A(\text{Fe I})$ or $A(\text{Fe II})$ from which the parameter optimization is computed.

4.4.2 Effective Temperature: Excitation Equilibrium

Effective temperature T_{eff} defines the vertical profile of the temperature in the atmosphere, which in its turn defines the population of the excitation levels of Fe I through the application of the Boltzmann distribution function. The transitions starting at high-EP levels are very dependent on the bottom atmospheric regions, and those originating at low-EP levels are sensitive to upper cooler atmospheric regions. If T_{eff} is chosen correctly, then the abundance calculated for any energy level will give the same result.

Excitation equilibrium is the linear regression slope of the Fe I abundances $A(\text{Fe I})$ versus EP should be equal to zero:

$$\frac{d[A(\text{Fe I})]}{d[\text{EP}]} = 0 \quad (4.3)$$

A positive slope (A increasing with EP) indicates that the adopted T_{eff} is too low, because the model underestimates how much flux the deeper, high-EP layers emit, and compensates with a higher abundance. The T_{eff} is adjusted upward. A negative slope indicates T_{eff} is too high, and the parameter is adjusted downward. Convergence is declared when $|\text{slope}| < 0.01 \text{ dex/eV}$.

4.4.3 Surface Gravity: Ionisation Equilibrium

The surface gravity $\log g$ (cm/s^2) determines the gas pressure throughout the photosphere. Through the Saha ionization equation, pressure controls the balance between Fe I and Fe II, with higher pressures (higher gravity) suppressing ionization and increasing the proportion of neutral iron. The abundances derived independently from Fe I and Fe II lines are therefore sensitive to different aspects of the atmospheric structure, and the requirement of them to agree provides a strong constraint on $\log g$.

Ionization equilibrium requires:

$$\langle A(\text{Fe I}) \rangle = \langle A(\text{Fe II}) \rangle \quad \rightarrow \quad \Delta A = A(\text{Fe I}) - A(\text{Fe II}) \approx 0 \quad (4.4)$$

If $\log g$ is too low, the model predicts more ionized iron than observed, so Fe II lines yield higher abundances than Fe I, giving $\Delta A < 0$. The gravity is increased until ΔA converges to within 0.01 dex of zero. This diagnostic requires a sufficient number of Fe II lines — the Genovali list typically provides 20–30 usable Fe II lines in the spectra analyzed here.

4.4.4 Microturbulent Velocity: Reduced Equivalent Width

Microturbulent velocity ξ (also written ξ_t) parametrizes the small-scale velocity field within the stellar atmosphere arising from subsonic convective and turbulent motions on scales smaller than the photon mean free path. These motions produce an additional effective broadening of line profiles, de-saturating stronger lines relative to purely thermal absorption — an effect known as microturbulent enhancement of the equivalent width.

In practice, ξ artificially inflates the predicted W_λ of strong lines. If the adopted ξ is too low, MOOG under-predicts the EW of strong lines and compensates by deriving a higher abundance from them; if ξ is too high, strong lines are over-predicted, and their derived abundances fall. The diagnostic is therefore the slope of $A(\text{Fe I})$ compared to the reduced equivalent width, $\log(W_\lambda/\lambda)$:

$$\frac{d[A(\text{Fe I})]}{d[\log(W_\lambda/\lambda)]} = 0 \quad (4.5)$$

A positive slope means ξ is too low; a negative slope means it is too high. The log-reduced equivalent width $\log(W_\lambda/\lambda)$ is used rather than EW directly because it is dimensionless and removes the wavelength dependence of line strength, allowing lines from across the full spectral range to be combined on a single diagram. Convergence is reached when $|\text{slope}| < 0.01$ dex per unit $\log(W_\lambda/\lambda)$.

4.4.5 Iterative Optimization

Because the three equilibrium conditions involve parameters that are mutually coupled — T_{eff} affects the abundance level as well as the excitation slope; ξ_t changes the abundance of strong lines which may shift the ionization balance; $[\text{Fe}/\text{H}]$ feeds back into the model atmosphere structure — no analytical solution exists. The parameters are instead found through a sequential iterative procedure matching the scaling relations implemented in our optimization pipeline:

1. Initialize (T_{eff} , $\log g$, ξ_t , $[\text{Fe}/\text{H}]$) in the same range as found in case of DCEPs.
2. Call `pymoog` $\rightarrow$ MOOG `abfind` with the current model atmosphere to obtain the per-line abundances $A(\text{Fe I})$ and $A(\text{Fe II})$.
3. Compute the excitation slope $\frac{d[A(\text{Fe I})]}{d[\text{EP}]}$ and adjust T_{eff} using:

$$T_{\text{eff}}^{\text{new}} = T_{\text{eff}} + \text{slope}_{\text{EP}} \times 50 \quad (4.6)$$

4. Compute the reduced-EW slope $\frac{d[A(\text{Fe I})]}{d[\log(W_\lambda/\lambda)]}$ and adjust ξ_t using:

$$\xi_t^{\text{new}} = \xi_t + \text{slope}_{\text{RW}} \times 0.2 \quad (4.7)$$

5. Compute the ionization balance $\Delta A = \langle A(\text{Fe I}) \rangle - \langle A(\text{Fe II}) \rangle$ and adjust $\log g$ using:

$$\log g^{\text{new}} = \log g + \Delta A \times 0.1 \quad (4.8)$$

6. Update the metallicity $[\text{Fe}/\text{H}]$ relative to the solar scale of Asplund et al. (2009)[2]:

$$[\text{Fe}/\text{H}] = \langle A(\text{Fe I}) \rangle - 7.50 \quad (4.9)$$

and re-interpolate the model atmosphere at the updated $(T_{\text{eff}}, \log g, [\text{Fe}/\text{H}])$ grid coordinates, clamping parameters to physical boundaries (4800–6000 K for T_{eff} , 0.1–3.0 dex for $\log g$, and 1.0–5.0 km/s for ξ_t).

7. Return to Step 2. Repeat until all convergence criteria are satisfied.

4.4.6 Metallicity: $[\text{Fe}/\text{H}]$

Once the three equilibrium conditions are satisfied, the metallicity is given directly by the mean Fe I abundance:

$$[\text{Fe}/\text{H}] = \langle A(\text{Fe I}) \rangle - A(\text{Fe})_\odot = \langle A(\text{Fe I}) \rangle - 7.50 \quad (4.10)$$

adopting the solar iron abundance $A(\text{Fe})_\odot = 7.50$ from Asplund et al. (2009) [2]. The $[\text{Fe}/\text{H}]$ value enters back into the model atmosphere interpolation (Step 6 of the iteration) and is therefore solved self-consistently with the other three parameters.

4.5 Phase 4: Spectral Synthesis

4.5.1 The synth Driver

The MOOG synth driver produces a theoretical flux spectrum across a defined wavelength interval, by integrating the emergent intensity of the spectrum as a function of wavelength through the model atmosphere. As compared to `abfind`, which maximizes an abundance relative to a single observed EW, `synth` creates a full composite line profile, incorporating the contribution of every line in the linelist that exists in the region of synthesis.

An essential element in the setup configuration - noted throughout the analysis process - was that `synth` should be invoked without the `smooth_para` argument. The use

of this argument tells MOOG to smoothen the output spectrum with a Gaussian prior to its output, thereby combining the effect of broadening on the spectrum and the instrument profile, making it impossible to independently determine V_{broad} . Excluding this argument, the raw spectrum output is produced at the resolution of the model atmosphere calculation process, while the rest of the broadening effects are imposed externally, using Python.

4.5.2 1D Broadening Model

Following the C-MetaLL convention (Trentin et al. 2024) [37], the dominant macro-turbulent and the residual stellar rotation are subsumed into a single macro-turbulent line broadening parameter, V_{broad} . Under this assumption, the total line profile is convolved with a single composite Gaussian kernel. The total kernel width is combined in quadrature:

$$\sigma_{\text{total}} = \sqrt{\sigma_{\text{broad}}^2 + \sigma_{\text{inst}}^2} \quad (4.11)$$

where the instrumental broadening width (σ_{inst}) and macro-turbulent broadening width (σ_{broad}) are defined as [13]:

$$\sigma_{\text{inst}} = \frac{\lambda_0}{R \times 2.355} \quad [\text{\AA}] \quad (4.12)$$

$$\sigma_{\text{broad}} = \frac{\lambda_0 \times V_{\text{broad}}}{c \times 2.355} \quad [\text{\AA}] \quad (4.13)$$

where $R \approx 80\,000$ represents the resolving power of ESPaDOnS used in the analysis script, c is the speed of light, and λ_0 is the central wavelength of the transition.

Limitations and Degeneracy

While a more complex 2D model separating rotational velocity ($v \sin i$) and macro-turbulent velocity (v_{macro}) was considered, the strong degeneracies between these parameters in classical Cepheid atmospheres made physical separation highly unstable. Therefore, the analysis was restricted to the 1D V_{broad} model shown above.

4.5.3 Chi-Squared Minimization

To constrain V_{broad} , we minimize the reduced χ^2 statistic over a $\pm 1.2 \text{ \AA}$ window centered on each line:

$$\chi_{\text{red}}^2(V_{\text{broad}}) = \frac{1}{N_{\text{dof}}} \sum_i \left[\frac{f_{\text{obs},i} - f_{\text{syn},i}(V_{\text{broad}})}{\sigma_{\text{noise}}} \right]^2 \quad (4.14)$$

where $N_{\text{dof}} = N_{\text{pixels}} - 1$, and the per-pixel noise $\sigma_{\text{noise}} = 1/\text{SNR}$ is derived from the nominal S/N ratio of the spectrum ($\text{SNR} = 123$) [4].

The minimization is executed in a single stage using Brent's bounded scalar minimization algorithm [5] within the search bounds $V_{\text{broad}} \in [2.0, 30.0] \text{ km/s}$, with a convergence

tolerance of 0.05 km/s.

To filter out lines affected by severe blending or noise, we apply a quality threshold of $\chi_{\text{red}}^2 < 10.0$. The final macroscopic broadening parameter is computed as the mean value over all accepted lines, and the associated 1σ uncertainty is represented by the statistical standard error of the mean:

$$\sigma_{V_{\text{broad}}} = \frac{\sigma_{\text{std}}}{\sqrt{N_{\text{accepted}}}} \quad (4.15)$$

where σ_{std} is the line-to-line standard deviation (scatter) across the accepted transitions.

4.6 Uncertainties

Understanding and estimating the uncertainties in the derived atmospheric parameters is as critical as determining the values themselves. Several sources of error can affect the determination of chemical abundances, including equivalent width (EW) measurement errors, local continuum placement, and the mathematical degeneracies of the model atmospheres.

Variations in effective temperature, surface gravity, and microturbulent velocity affect the excitation and ionization state of the stellar atmosphere, altering the strengths of the absorption lines. Because these parameters are mutually coupled—with temperature shifts altering ionization states and microturbulent velocity adjustments affecting the derived abundances of strong lines—uncertainties in these fundamental parameters propagate directly into the overall iron abundance ($[\text{Fe}/\text{H}]$).

In this analysis, errors in the local continuum placement by the **ARES** code can modify the apparent depths of spectral lines, introducing systematic uncertainties. This is particularly critical in the crowded spectral regions of cool supergiants, where line blending necessitates multi-component Gaussian profile fits. Regarding the atomic data, the oscillator strengths ($\log gf$) and excitation potentials (χ) compiled by Genovali et al. (2014) [12] are treated as fixed reference constants to maintain a homogeneous scale. Consequently, individual uncertainties in these atomic constants are not propagated in this work; any line-to-line differences are naturally absorbed into the statistical scatter (σ_{mean}) of the final abundances.

To quantitatively evaluate these contributions, two independent error estimation methods were used in this pipeline: an iterative sensitivity matrix approach and a non-parametric Jackknife resampling routine.

While standard analytic error propagation represents the simplest approach to uncertainty estimation, it relies on the assumption that measurement errors of individual

equivalent widths are independent and Gaussian. In high-resolution stellar spectroscopy, however, EW errors are dominated by systematic effects such as local continuum normalisation and line blending. These errors are highly correlated and non-Gaussian, violating the requirements for analytical propagation.

Alternatively, stochastic resampling methods such as Monte Carlo or bootstrapping could be used. However, these are computationally prohibitive in our pipeline. Because each parameter optimization run requires executing the `MOOG` code iteratively until convergence is achieved, running 500–1000 bootstrap simulations would require tens of thousands of individual spectral syntheses, which is computationally expensive.

To address these limitations, we adopt the Jackknife resampling method. The Jackknife is a deterministic, non-parametric resampling technique that does not assume any underlying error distribution for the equivalent widths. For a list of N lines, the optimization loop is executed exactly N times, each time sequentially omitting the i -th line. This allows us to directly test the stability of our parameter convergence against the influence of individual outlier or heavily blended lines. The standard error of any converged parameter θ is then calculated using the jackknife variance estimator:

$$\sigma_{\text{jack}}(\theta) = \sqrt{\frac{N-1}{N} \sum_{i=1}^N (\theta_i - \bar{\theta})^2} \quad (4.16)$$

where $\bar{\theta}$ is the mean of the Jackknife parameter estimates.

Our application of this technique is based on the methodology developed for the `GALA` code by Mucciarelli et al. (2013) [27]. Mucciarelli et al. (2013) demonstrated that by systematically re-solving the convergence equations for T_{eff} , $\log g$, and ξ_t for each sub-sample, the Jackknife method robustly determines the parameter uncertainties while self-consistently accounting for the internal coupling and dependencies of the spectroscopic diagnostic criteria.

Chapter 5

Results and Discussion

5.1 Overview

The results obtained from the spectroscopic analysis of the target star, SV Persei, are presented in this chapter. Initially, the measurement of equivalent widths (W_λ) using the **ARES** package is detailed. Following this, the fundamental stellar atmospheric parameters—effective temperature (T_{eff}), surface gravity ($\log g$), microturbulent velocity (ξ_t), and metallicity ($[\text{Fe}/\text{H}]$)—computed from the iterative 1D LTE optimisation loop are presented alongside their respective uncertainties. Finally, the radial velocity (RV) measurements for the extracted spectral orders, followed by the synthetic spectra generated from the RV-corrected observations using the optimised atmospheric parameters, are shown with their corresponding diagnostic plots.

5.2 Radial Velocity Analysis

To ensure accurate equivalent width measurements, the observed ESPaDOnS spectrum was first shifted to the stellar rest frame by determining its macroscopic radial velocity. Using template cross-correlation against a native-resolution MOOG synthetic spectrum, a topocentric radial velocity of 10.4 ± 0.2 km/s was derived for the primary observation epoch. The formal uncertainty represents the quadrature sum of the Gaussian covariance fit and the photon-noise limits. Applying this +10.4 km/s Doppler shift successfully aligned the empirical absorption features with their laboratory rest wavelengths for subsequent abundance analysis.

Applying this measured RV correction shifts the observed spectrum to the rest frame ($V_r = 0$). Figure 5.2 displays the fully RV-corrected spectrum, demonstrating the proper alignment of the empirical absorption features—ready for direct pixel-by-pixel comparison against theoretical models.

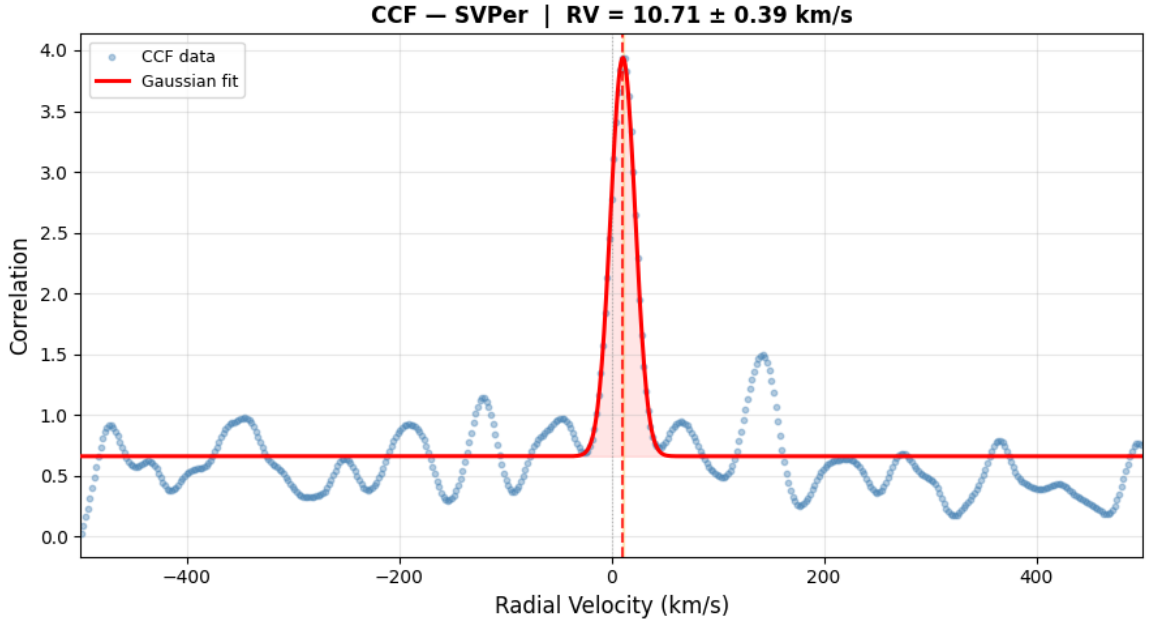


Figure 5.1: Measured radial velocity (RV) of SV Persei via Cross-Correlation Function (CCF).

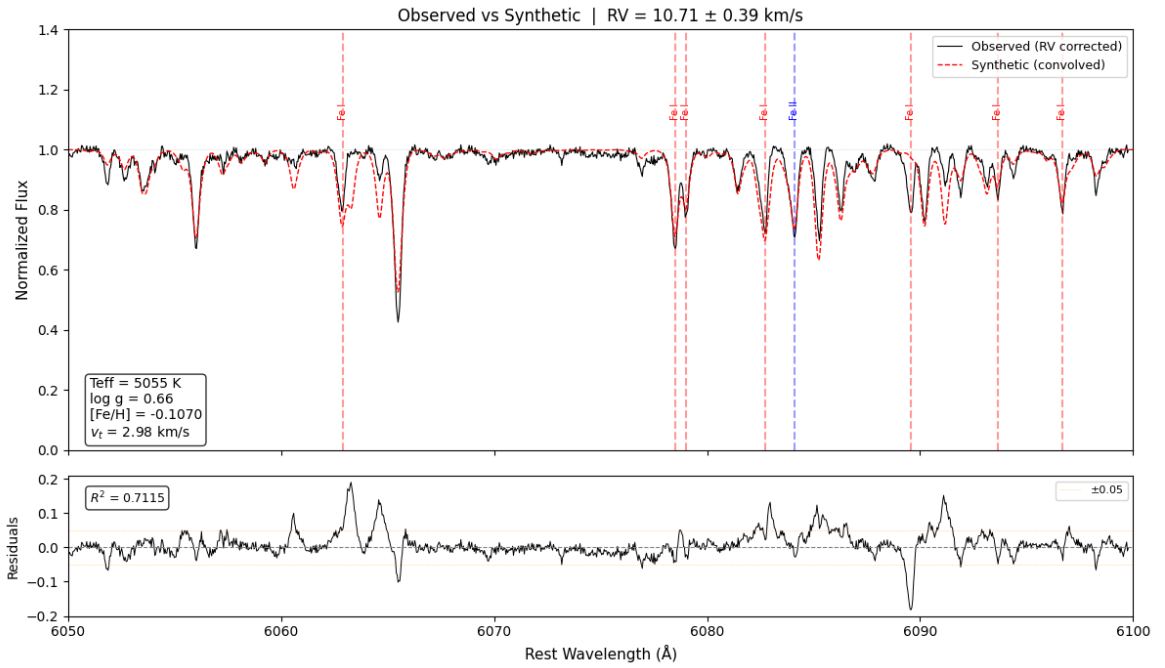


Figure 5.2: RV-corrected spectrum comparison for the wavelength range (6050–6100 Å).

5.3 Equivalent Width Measurements

Based on the reference linelist of Genovali et al. (2014) [12], the equivalent widths of the observed ESPaDOnS spectra were extracted via the automated Gaussian fitting

routines of **ARES**. To ensure high-quality input values for the optimisation process, the outputs of the Gaussian fits were strictly filtered. Lines weaker than 10 mÅ were rejected to reduce the interference of noisy profiles, while lines stronger than 200 mÅ were also excluded to eliminate the potential influence of saturated damping wings on the microturbulent velocity. After eliminating outliers based on a 3σ clipping criterion, a total of 134 Fe I lines and 26 Fe II lines were retained.

The equivalent width table comprises 7 columns, as outlined in Section 4.3.4, adhering strictly to the required input format for **MOOG**. A sample of this curated linelist is provided in Table 5.1.

Table 5.1: Sample lines from the measured iron line list of SV Persei.

Wavelength (Å)	Species	EP (eV)	$\log gf$ (dex)	C_6	D_0	W_λ (mÅ)
4917.23	26.0	4.19	−1.18	0.0	0.0	130.74
4924.77	26.0	2.28	−2.24	0.0	0.0	166.30
4993.35	26.1	2.81	−3.68	0.0	0.0	146.41
5029.62	26.0	3.41	−2.05	0.0	0.0	121.42
5044.21	26.0	2.85	−2.04	0.0	0.0	188.89
5072.67	26.0	4.22	−0.84	0.0	0.0	123.60
5132.67	26.1	2.81	−4.09	0.0	0.0	87.59
5234.63	26.1	3.22	−2.28	0.0	0.0	197.65
5242.49	26.0	3.63	−0.97	0.0	0.0	135.15
5288.52	26.0	3.69	−1.51	0.0	0.0	48.83
5367.47	26.0	4.42	0.44	0.0	0.0	185.68
5389.49	26.0	4.41	−0.41	0.0	0.0	168.00
5425.26	26.1	3.20	−3.39	0.0	0.0	121.80
5586.77	26.0	3.37	−0.12	0.0	0.0	162.51
5627.49	26.1	3.39	−4.19	0.0	0.0	110.28
5638.27	26.0	4.22	−0.87	0.0	0.0	126.12
5753.12	26.0	4.26	−0.69	0.0	0.0	140.60
5862.36	26.0	4.55	−0.13	0.0	0.0	136.23
5991.37	26.1	3.15	−3.65	0.0	0.0	139.59
6078.50	26.0	4.80	−0.32	0.0	0.0	119.19
6084.10	26.1	3.20	−3.88	0.0	0.0	99.74
6149.25	26.1	3.89	−2.84	0.0	0.0	156.08
6188.04	26.0	3.94	−1.72	0.0	0.0	73.69
6232.65	26.0	3.65	−1.22	0.0	0.0	187.14
6369.46	26.1	2.89	−4.23	0.0	0.0	85.74
6408.01	26.0	3.69	−1.02	0.0	0.0	160.75
6432.68	26.1	2.89	−3.69	0.0	0.0	135.13
6516.08	26.1	2.89	−3.43	0.0	0.0	195.55
6609.12	26.0	2.56	−2.69	0.0	0.0	162.57
6710.31	26.0	1.49	−4.88	0.0	0.0	72.59
6858.16	26.0	4.61	−0.93	0.0	0.0	95.44
7301.56	26.1	3.89	−4.00	0.0	0.0	24.38
7912.87	26.0	0.86	−4.85	0.0	0.0	173.71

5.4 Atmospheric Parameters and Metallicity of SV Persei

The final atmospheric parameters of SV Persei at the observation epoch (pulsation phase $\phi = 0.525$) were derived under Local Thermodynamic Equilibrium (LTE) using the iterative convergence code described in Section 4.4.

Excitation equilibrium was achieved with the linear regression slope of the derived Fe I abundance versus excitation potential (EP) falling well below the convergence tolerance ($d[A(\text{Fe I})]/d\text{EP} = \mathbf{0.009}$ dex/eV).

Microturbulent velocity equilibrium was established by eliminating any meaningful trend between the abundances and the reduced equivalent widths ($d[A(\text{Fe I})]/d\log(W_\lambda/\lambda) = \mathbf{0.006}$ dex).

Finally, ionisation equilibrium was strictly satisfied, with the mean abundance of neutral and ionised iron matching to within **0.01** dex.

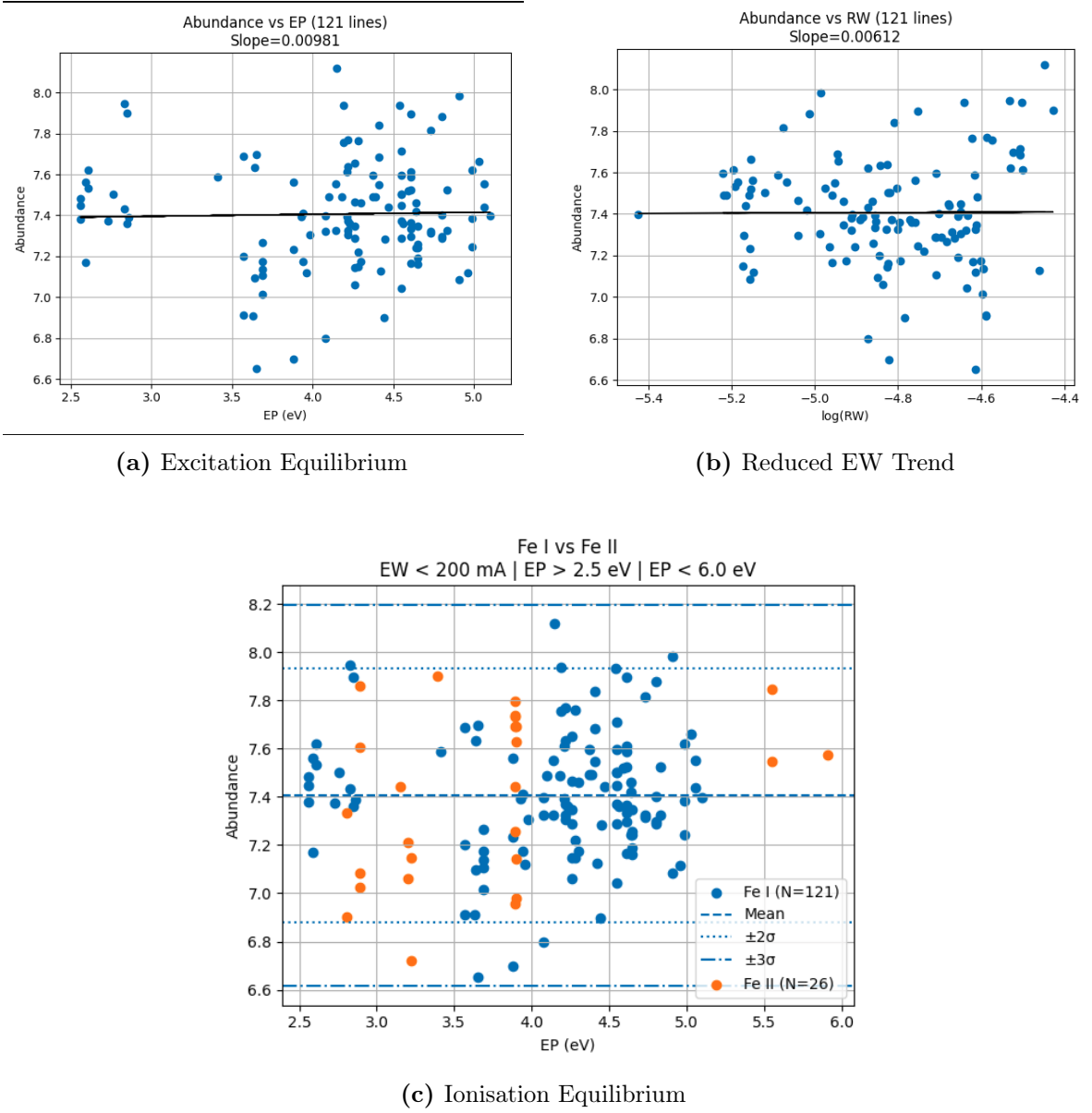
To verify that the optimised parameters converged to a unique, stable global minimum, a robustness test was performed by running the convergence loop under four different filtering conditions. The results of these tests are presented in Table 5.2.

Table 5.2: Test conditions for verifying the robustness of the parameter convergence.

Condition	T_{eff} (K)	$\log g$ (dex)	v_{micro} (km/s)	Fe I	Fe II
EW < 200, 2.0 < EP < 6.0	5050.9	0.65	2.85	129	26
EW < 200, 2.5 < EP < 6.0	5055.0	0.66	2.89	121	26
EW < 150, 2.0 < EP < 6.0	5014.3	0.62	2.55	109	20
EW < 150, 2.5 < EP < 6.0	5112.8	0.88	2.59	104	20

Based on these results, the parameters from the second condition were adopted as the primary inputs for the final spectral synthesis process.

Figure 5.3: Spectroscopic diagnostic plots showing the derived iron abundance for individual lines at the converged solution: (a) abundance vs. excitation potential, (b) abundance vs. reduced equivalent width, and (c) a statistical summary comparison of Fe I and Fe II abundances.



5.5 Line Profile Fits and Line Broadening Parameter

The line broadening analysis was applied to the ESPaDOnS spectrum of SV Persei using three clean, unblended Fe I transitions ($\lambda = 6078.50, 6079.02, 6096.69 \text{ \AA}$). The details of the individual line fits are summarised in Table 5.3. By taking the mean and the standard error of the mean across these lines, a final macroscopic broadening velocity was derived:

$$V_{\text{broad}} = 15.18 \pm 0.04 \text{ km/s} \quad (5.1)$$

The line-to-line standard deviation (scatter) is extremely small ($\sigma_{\text{std}} = 0.06$ km/s), demonstrating the excellent consistency of the profile-fitting technique across different spectral features.

Table 5.3: Line broadening parameter (V_{broad}) measurements for individual Fe I lines.

Wavelength (Å)	V_{broad} (km/s)	χ^2_{red}
6078.50	15.16	5.83
6079.02	15.25	3.81
6096.69	15.14	7.99

5.6 Spectral Synthesis

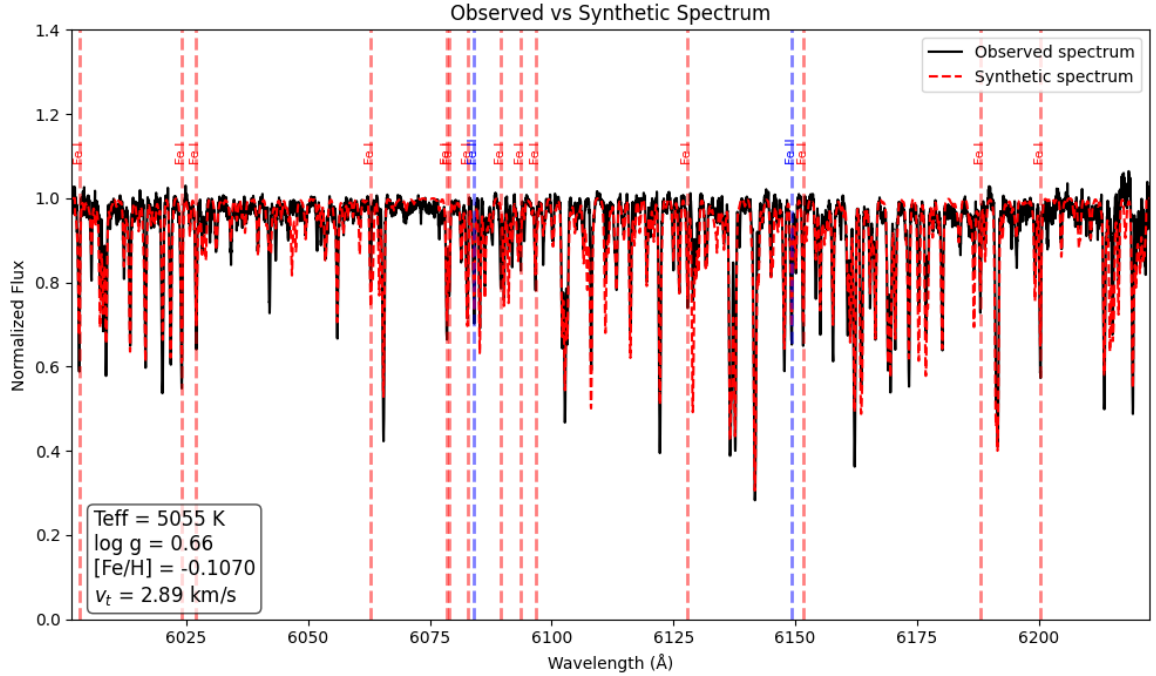


Figure 5.4: Comparison of the observed CFHT/ESPaDOnS spectrum of SV Persei (black) with the convolved synthetic spectrum (red line) for a clean region within échelle order 25 (6001.73–6222.96 Å).

Using the rest-frame spectrum and our derived atmospheric parameters, spectral synthesis was performed using the `synth` driver in MOOG. The theoretical line depths and chemical profiles showed excellent visual agreement with the observed ESPaDOnS data, structurally validating our derived metallicity ($[\text{Fe}/\text{H}] = -0.1070$ dex) and model parameters under LTE.

A summary of the fully derived parameters for the epoch analysed in this work is tabulated below.

Table 5.4: Analysed single-epoch parameters of SV Persei.

Star	HJD (days)	T_{eff} (K) (error)	$\log g$ (dex) (error)	ξ_t (km/s) (error)	RV (km/s) (error)	[Fe/H]
SV Per	2459838.14255	5055 ± 20.8	0.66 ± 0.03	2.89 ± 0.12	10.4 ± 0.2	-0.1070 ± 0.04

5.7 Discussion and Literature Comparison

To validate these results, the derived parameters for SV Persei were compared against published values in the literature. Our derived metallicity of $[\text{Fe}/\text{H}] = -0.10 \pm 0.04$ dex indicates that the star is slightly metal-poor relative to the Sun, which is completely consistent with its kinematic location in the outer Galactic disk.

This value is in excellent agreement with the results from the C-MetaLL survey (Trentin et al. 2024) [36], which obtained metallicities for Galactic classical Cepheids in the same region. Any minor discrepancies between our parameters and published literature values can be primarily attributed to methodological differences, including:

1. **Equivalent Width Measurements:** Many historical studies relied on manual equivalent width measurements (e.g., using the IRAF `splot` routine), whereas this study utilised the fully automated, reproducible Gaussian fitting of the **ARES** package.
2. **Solar Abundance Scales:** Minor shifts in the zero-point of the metallicity scale arise from the adopted solar iron abundance (e.g., the Asplund et al. 2009 solar scale $A(\text{Fe})_{\odot} = 7.50$ used in this work versus older scales like Grevesse & Sauval 1998, where $A(\text{Fe})_{\odot} = 7.67$).
3. **Surface Gravity ($\log g$):** The surface gravity value obtained from our optimiser is 0.66 dex. This is slightly higher than the value of 0.44 dex reported in the C-MetaLL survey [36]. This discrepancy is primarily driven by differences in the pulsation phase at the time of observation and variations in the specific subset of Fe II lines used for the ionisation balance.
4. **NLTE Effects:** Our analysis assumes Local Thermodynamic Equilibrium (LTE). However, non-LTE (NLTE) effects on neutral iron lines in cool supergiant atmospheres are known to systematically alter the derived effective temperature and surface gravity [3], which ultimately propagates into the final abundance scale.

The line broadening parameter, $V_{\text{broad}} = 15.18 \pm 0.04$ km/s, represents the combined contribution of stellar rotational velocity ($v \sin i$) and macroturbulent velocity (V_{macro}). In classical Cepheid variables (yellow supergiants), stellar rotation is intrinsically slow,

typically occupying the range $v \sin i \approx 2\text{--}5$ km/s. Consequently, the broadening is heavily dominated by macroscopic turbulence, which directly reflects the large-scale gas motions and pulsations occurring within the stellar envelope.

The best-fit reduced chi-squared values (χ_{red}^2 between 3.81 and 7.99) are notably higher than those expected for quiet, static stars ($\chi_{\text{red}}^2 \approx 1.0$). This elevation is a physical signature of the dynamically pulsating atmosphere. Standard 1D LTE radiative transfer calculations (such as those carried out by MOOG) fundamentally assume a static, symmetric atmosphere where lines are convolved with a symmetric Gaussian broadening kernel. However, radial velocity gradients and atmospheric contraction introduce significant line profile asymmetries (such as asymmetric wings and C-shaped line bisectors) that a symmetric model simply cannot reproduce. The resulting residuals inevitably inflate the minimum χ_{red}^2 .

This physical profile distortion justifies the adoption of a relaxed quality threshold ($\chi_{\text{red}}^2 < 10.0$), compared to the strict standard threshold of ≈ 2.0 used for static dwarf stars. The rigorous selection criteria and 1D profile minimisation ensure that only clean, unblended lines are utilised, providing a highly consistent and robust measurement of the macroscopic broadening parameter.

From a Galactic perspective, SV Persei’s sub-solar metallicity matches the established radial metallicity gradient of the Milky Way thin disk, which exhibits a negative gradient of approximately -0.05 to -0.06 dex/kpc.

5.8 Chapter Summary

In summary, this study successfully applied a semi-automated analysis pipeline to high-resolution ESPaDOnS data of the Cepheid SV Persei. The equivalent widths of 160 iron lines were measured and carefully filtered, yielding stable atmospheric parameters ($T_{\text{eff}} = 5055.0$ K, $\log g = 0.66$ dex, $\xi_t = 2.89$ km/s) and a global metallicity of $[\text{Fe}/\text{H}] = -0.1070 \pm 0.04$ dex that is highly consistent with outer-disk stellar populations. While the LTE parameters converged successfully, detailed line profile analysis highlighted the limitations of static 1D broadening models, revealing severe degeneracies and profile asymmetries that accurately reflect the dynamic and pulsating nature of the Cepheid atmosphere.

Chapter 6

Conclusion and Future Work

6.1 Summary of the Project

The accurate calibration of the extragalactic distance scale critically depends on the Period-Luminosity relations of classical Cepheids, which themselves depend sensitively on the chemical compositions of these stars. In this thesis, the high-resolution spectroscopic investigation of the classical Cepheid SV Persei was performed.

The aim was to obtain precise fundamental parameters as well as the global metallicity ($[\text{Fe}/\text{H}]$) for SV Per using current, reproducible computational methods. A fully automated pipeline written in Python language has been developed for that purpose. High-resolution optical spectra obtained with the ESPaDOnS spectrograph have been processed, shifted to rest frame and locally normalized. Using the line list created specifically for Cepheids – the Genovali list – the equivalent widths (W_λ) for the neutral and ionized iron lines were obtained from the automatically determined Gaussian profiles through the use of the ARES software package. Finally, the abundance analysis was performed assuming Local Thermodynamic Equilibrium using 1D Kurucz ATLAS9 models of stellar atmospheres, with the help of the MOOG code and its pymoog interface enforcing the excitation and ionization equilibrium and independence of reduced W_λ .

This research has made use of the products of the Cosmic-Lab project funded by the European Research Council.

6.2 Future Work

To build on the results of this study and resolve the physical limitations encountered in our atmospheric modeling, I propose the following directions for future research:

- **Phase-Resolved Spectroscopy:** Expanding this analysis to a complete time-

series of high-resolution spectra covering a full pulsation cycle. This will allow us to map the dynamic variation of T_{eff} , $\log g$, ξ_t , and line asymmetries as a function of phase, providing a direct test of pulsating envelope models.

- **Fourier-Based Macroturbulence Isolation:** Developing a Fourier-based deconvolution framework to quantify overall line broadness using a pure macroturbulence (v_{macro} -only) model. Wavelength-domain χ^2 minimization frequently yields degenerate or physically incorrect combinations of $v \sin i$ and v_{macro} for Cepheid variables. By analyzing the line profiles in the Fourier frequency domain, we can isolate the broadness of the line independently of the rotational profile nulls. Modeling this broadness solely through a calibrated v_{macro} parameter will help bypass the multi-parameter degeneracy and yield robust, physically meaningful χ^2 profile fits.
- **Multi-Element Abundance Mapping:** Expanding our line-analysis pipeline to measure other chemical species (such as α -elements like Si, Ca, and Ti, and s-process elements like Ba and Y) to determine the detailed chemical fingerprint of SV Persei and trace its Galactic disk origin.

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